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Towards Universal AI-Generated Image Detection by Variational Information Bottleneck Network

Haifeng Zhang, Qinghui He, Xiuli Bi, Weisheng Li, Bo Liu*, Bin Xiao
Chongqing University of Posts and Telecommunications
s220331143@stu.cqupt.edu.cn, s220232014@stu.cqupt.edu.cn
{bixl, liws, boliu, xiaobin}@cqupt.edu.cn

Abstract

The rapid advancement of generative models has significantly improved the quality of generated images. Meanwhile, it challenges information authenticity and credibility. Current generated image detection methods based on large-scale pre-trained multimodal models have achieved impressive results. Although these models provide abundant features, the authentication task-related features are often submerged. Consequently, those authentication task-irrelated features cause models to learn superficial biases, thereby harming their generalization performance across different model genera (e.g., GANs and Diffusion Models). To this end, we proposed VIB-Net, which uses Variational Information Bottlenecks to enforce authentication task-related feature learning. We tested and analyzed the proposed method and existing methods on samples generated by 17 different generative models. Compared to SOTA methods, VIB-Net achieved a 5.55% improvement in mAP and a 9.33% increase in accuracy. Notably, in generalization tests on unseen generative models from different series, VIB-Net improved mAP by 12.48% and accuracy by 23.59% over SOTA methods. The code is available at https://github.com/oceanzhf/VIBAIGCDetect.

1. Introduction

The continuous advancements in deep learning have led to the emergence of numerous well-known image generation models, such as GANs[9][15] and Diffusion[12][31]. Images generated by these models have reached a level where observers can hardly distinguish them from real images, and the flexibility of tools based on these models has also greatly improved [6]. With a few simple steps, people can easily obtain high-quality images as needed. While this presents a tremendous opportunity for fields like visual arts, it also serves as a paradise for individuals looking to fabricate misinformation. Given the ease of creating manipulated images today and the potential risk of using them for malicious purposes, there is an urgent need to develop detectors that help distinguish between real and generated images.

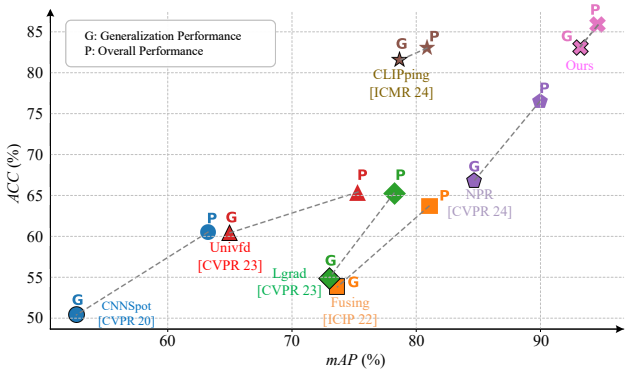


Figure 1. Each method's overall performance('P') on test data and generalization performance ('G') on unseen data are linked by dashed lines. Our detector demonstrates a clear advantage over all SOTA methods in both metrics.

Existing detection methods [4][39][25][43] focus on identifying specific traces left by generators in either the spatial or frequency domain of generated images, thus establishing a decision boundary between real and generated images. These detection methods are highly effective in detecting images generated by GANs and earlier diffusion models. However, their performance drops significantly when facing images generated by recent, popular models that visually resemble real images. Although some error reconstruction-based methods [40][23] have been developed specifically for detecting images generated by diffusion models, they similarly experience performance declines when applied to GAN-generated images. Consequently, current research increasingly focuses on improving the cross-model genera detection capabilities of detectors.

Generalization across different domains is a primary re-

* Corresponding Author

search challenge in deep learning. In this context, the emergence of large-scale pre-trained multimodal models has introduced new approaches and promising results. For instance, by using the image encoder from the CLIP [30] model to extract image features and training a classifier with linear probing, there is a notable improvement in generalization performance over models like ResNet50[28][6]. However, compared to traditional classification tasks, real and generated images share subtle, harder-to-capture distinguishing features. This implies that the features extracted by large-scale pre-trained models for the task of distinguishing real from fake images may contain a considerable amount of task-irrelevant features. Task-irrelevant features can either have no impact on the realness of the image or, though effective in isolation, become redundant when combined with other features. These features not only interfere with the classifier's ability to accurately learn decision boundaries but also hinder the model's ability to capture and distinguish task-relevant characteristics. More critically, task-irrelevant features may lead the model to learn superficial biases. For example, if a fake image contains a tree while the real image does not, the model might erroneously associate the presence of the tree with being a fake image. Consequently, when detecting other images, the model may mistakenly classify a real image as fake if it contains a tree. Therefore, when dealing with high-quality generated images with subtle distinguishing features, it is challenging to fine-tune the model to separate essential from unnecessary information. This can lead to an overfitting of the incidental statistical correlations between unnecessary information and target images, ultimately undermining the model's generalization and discrimination capabilities.

How can we harness the powerful feature extraction capability of large models while minimizing the interference of unnecessary information? By approaching this from a novel perspective, we propose addressing the issue using Information Bottleneck Theory. This theory aims to create latent encodings that not only effectively represent the target but also maximally compress the original input, thereby avoiding the impact of task-irrelevant features on the model. Specifically, we introduce a method called VIB-Net, which couples a large-scale pre-trained multimodal model with Variational Information Bottleneck (VIB) [1]. Considering that feature representations extracted by the model's image encoder are more task-specific, we decouple and filter these features using latent variables. This method maximizes the mutual information between the latent feature and the classification output, while minimizing the mutual information between the latent feature and the input, resulting in a concise feature that should only contain the effective information needed for the task. This concise latent feature helps the model avoid overfitting to irrelevant features and the incidental correlations with the task, preventing the model from learning superficial biases, which is beneficial to the model's generalization performance.

In summary, our contributions are as follows: (1) To the best of our knowledge, we are the first to apply Information Bottleneck theory and techniques to the detection of generated images, demonstrating the potential of this theory and approach in this field. (2) We analyze the limitations of methods based on large-scale pre-trained models in detecting images generated by unseen generative models, and present our theory of what could be wrong with the existing paradigm. (3) We propose VIB-Net for the detection of generated images, which uses VIB to decouple and filter the general features extracted by large-scale pre-trained multimodal models, removing task-irrelevant information to enhance detection performance. (4) A large number of experiments validate the advantages of our method (see Fig. 1), especially in the improvement of generalization performance (+23.59% accuracy).

2. Related work

Generalization Performance. The continuous advancement of generated image models presents ongoing challenges for generated image detection, making the generalization performance of detection models on unseen generators an increasing focus for researchers[14][21]. [39] explored the effects of data preprocessing and augmentation methods on the generalization performance of a ResNet50[11]-based detector. [44] argues that the differences between real and fake images are subtle and localized, framing forgery detection as a fine-grained task that could positively impact generalization performance. [3] takes a frequency-domain approach, mapping real images to a dense subspace and synthetic images outside this subspace to enhance generalization. While these methods have achieved certain successes on different models within the same generation series, they show limitations across different series. This is due to the fact that, with the development of generative algorithms, subtle traces in generated images are becoming increasingly difficult to capture. Additionally, artifacts in the frequency domain differ between model series and are gradually decreasing in visibility.

The impressive results of large-scale pre-trained models in terms of transferability and generalization have sparked significant interest among researchers in the field of generated image detection. [28] argues that existing detectors perform asymmetric learning, where they use CLIP's image encoder to extract image features, exploring the potential of large models in detecting different model families. [6] suggests that detectors based on large models rely on higher-level semantic features, which enhances the generalization ability of the detectors. [19] uses both image and text features extracted by the CLIP model, and then employs a multilayer perceptron to detect images generated by

diffusion models, improving generalization performance.

Information Bottleneck. The Information Bottleneck (IB) [38] is an innovative information-theoretic method used for pattern analysis and representation learning. It provides an elegant balance between data compression and information retention[13]. VIB [1] is a variational approximation of IB, aimed at parameterizing IB using neural networks. [2] demonstrated VIB's capability in improving classification calibration and out-of-distribution data. [24] explored VIB as a regularization technique to address overfitting issues in NLP models. [41] proposed a dual-stream information bottleneck method to utilize in-distribution data to enhance out-of-distribution target detection tasks.

Based on the above analysis, current methods for detecting generated images using large-scale pre-trained models simply utilize the general representations extracted from images, without thoroughly exploring the relationship between these general features and downstream tasks. Additionally, Information Bottleneck is an effective technique for improving performance in cross-domain and cross-distribution tasks. Therefore, unlike existing methods, our approach leverages Information Bottleneck techniques to explore the potential of large-scale pre-trained models in generated image detection from a completely new perspective, achieving outstanding performance.

3. Preliminaries

3.1. Problem background

A common paradigm for using large-scale pretrained models is to leverage a large-scale pretrained multimodal model (e.g., CLIP) to learn the general representation of real and generated images, and then apply this feature representation to downstream tasks[28][6]. Building on their findings, we further consider the following points: (1) It is well-known that the features extracted by large-scale pre-trained visual encoders are general-purpose, meaning they contain many general features, such as semantic information. (2) Another widely accepted notion is that some critical features for distinguishing real from generated images are subtle and local.

Further combining these two points, we propose the following hypotheses: (i) The general features extracted by large-scale pre-trained multimodal models are more beneficial for real-fake image classification tasks than those extracted by pre-trained visual models. (ii) These general features contain task-irrelevant features for real-fake classification. (iii) These task-irrelevant features affect the classification of real and fake images. Next, we will delve into these three hypotheses.

3.2. Analysis of general feature representation

For the first hypothesis, We use image pairs of real and generated images to aid our analysis[6], generating textual descriptions for 1,000 real images and using these descriptions to produce 1,000 corresponding images with DALL·E 2 and DALL·E 3. We then compare features extracted by large-scale pre-trained multimodal models(CLIP[30]) and pre-trained visual models(ResNet[11]) through UMAP [26] visualization of 3,000 images. The results show that features extracted by multimodal models (e.g., CLIP) exhibit greater distribution differences for real images and DALL·E 2-generated images, suggesting multimodal models may be better at distinguishing real from generated images. However, for DALL·E 3-generated images, both model types show similar feature distributions, possibly due to advancements in image generation tools, which complicate the classification task and require further fine-tuning.

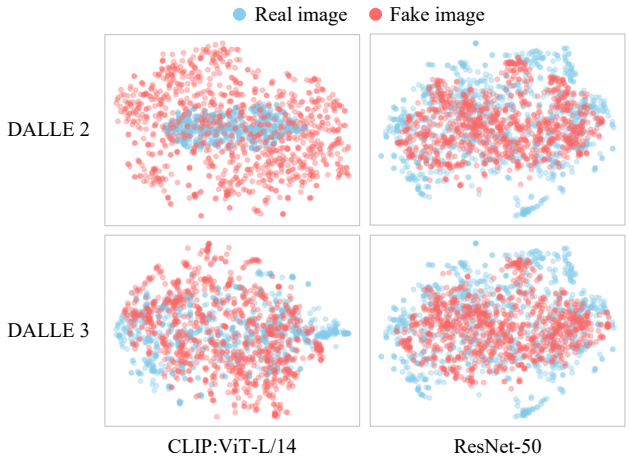
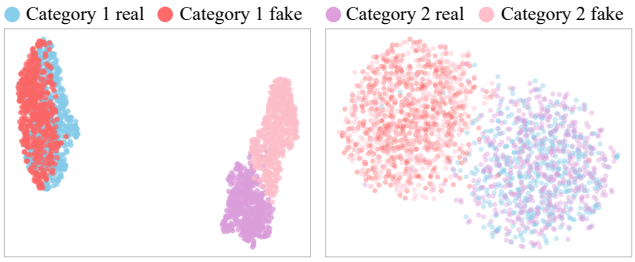


Figure 2. UMAP visualization of features extracted from Dalle2 and Dalle3 generated images using large-scale pretrained multimodal(CLIP) and unimodal models(ResNet).

To further explore hypotheses (ii) and (iii), we visualized the extracted features using both category and authenticity as labels to observe their distribution differences. Specifically, we prepared images of two categories (car and cat), each containing 500 real images and 500 generated images. We then used CLIP to extract their features and performed UMAP dimensionality reduction for visualization, with the results shown in Fig. 3a. We can see that the features extracted by CLIP clearly form two clusters: the left cluster mainly consists of light blue and light red points, while the right cluster is mainly composed of light pink and light purple points. These two clusters correspond to category 1 and category 2, respectively. This indicates that CLIP can significantly distinguish category information. However, this clear separation of category information may affect the decision-making process for classifying real vs. generated images. When examining the distinction between real and generated images, we notice that the right cluster shows a noticeable separation of real and generated features, with



(a) Visualization of features extracted by CLIP.

(b) Visualization of features extracted by VIB-Net.

Figure 3. UMAP visualization of features extracted from real and generated images of two category using CLIP and VIB-Net.

two different colors observed within the cluster. This suggests that CLIP is capable of distinguishing real and generated features within category 2. However, in the left cluster, CLIP does not show significant ability to differentiate between real and generated features. We observe that in the left cluster (category 1), the two colors are mixed together, showing poor distinguishability. From these observations, it is easy to infer that the internal separation of real or generated images caused by semantic category information makes it more difficult for the classifier to learn the decision boundary. More importantly, the strong sensitivity of large-scale pre-trained multimodal models like CLIP to semantic information leads the model to tend to learn superficial biases.

Based on this viewpoint, we consider whether it is possible to minimize the irrelevant features within these multimodal features while retaining the ones that are relevant to the classification task. This would help the model avoid learning superficial biases, allow the classifier to more easily learn decision boundaries, and thereby enhance the model's discrimination ability and generalization capability. This is the motivation of this paper. As shown in Fig. 3b, the method(VIB-Net) proposed in this paper achieves significant results.

4. Approach

Based on the issues analyzed in Sec. 3, an effective approach is to filter the multimodal features in order to obtain the representations most beneficial for the real-fake classification task, which serves as the starting point of this paper. Overall, our method is illustrated in Fig. 4. VIB attempts to filter out irrelevant features and to find the most concise representation, making it easier for the classifier to learn the features needed for distinguishing between real and fake tasks, thereby significantly improving the accuracy of the classification task. Below, we will provide a detailed description of our method.

4.1. Information bottleneck constraint

To specifically optimize the filtering of irrelevant information in the input features while retaining task-relevant information, we apply the information bottleneck (IB) principle as a constraint on the features to be optimized. Let X, Z, and Y represent the input, the latent compressed representation, and the output target, respectively. The goal of IB is to find the maximum compressed representation Z of the input X, while preserving as much information about the output Y as possible. The maximization objective function is defined as follows:

$$L_{IB} = I(Z,Y) - \beta I(Z,X) \tag{1}$$

Where $I(.,.)$ denotes mutual information and β is a trade-off parameter. The first term constrains the latent compressed representation to retain sufficient information for downstream tasks, while the second term ensures that irrelevant and redundant information is filtered out.

4.2. Variational information bottleneck

To parameterize the information bottleneck (IB) using deep neural networks, we perform variational inference on Equation (1). First, based on the definition of mutual information, Equation (1) can be rewritten as:

$$\begin{aligned} L_{IB} = & \int dydzp(y,z) \log \frac{p(y|z)}{p(y)} \\ & -\beta \int dxdzp(x,z) \log \frac{p(z|x)}{p(z)} \end{aligned} \tag{2}$$

Where $p(y|z)$ is defined by the encoder and the Markov chain as follows:

$$p(y|z) = \int dx \frac{p(y|x)\,p(z|x)\,p(x)}{p(z)} \tag{3}$$

Since it is not computable here, we consider setting $q(y|z)$ as a parameterized approximation of $p(y|z)$, serving as a "decoder," which can be approximated by a linear layer. We introduce the Kullback-Leibler (KL) divergence to obtain an optimizable representation. As the KL divergence satisfies $\mathrm{KL}(p(y|z), q(y|z)) \geq 0$, and by applying the Markov assumption, $p(y,z)$ can be rewritten as $p(y,z) = \int dxp(x)p(y|x)\,p(z|x)$, thereby providing a lower bound for the first term of the objective function:

$$I(Z,Y) \geq \int dxdydzp(x)p(y|x)\,p(z|x) \log q(y|z) \tag{4}$$

At the same time, let $r(z)$ be an estimate of the prior probability $p(z)$, used to approximate the marginal distribution, thereby obtaining an upper bound for the second term:

$$I(Z,X) \leq \int dxdzp(x)\,p(z|x) \log \frac{p(z|x)}{r(z)} \tag{5}$$

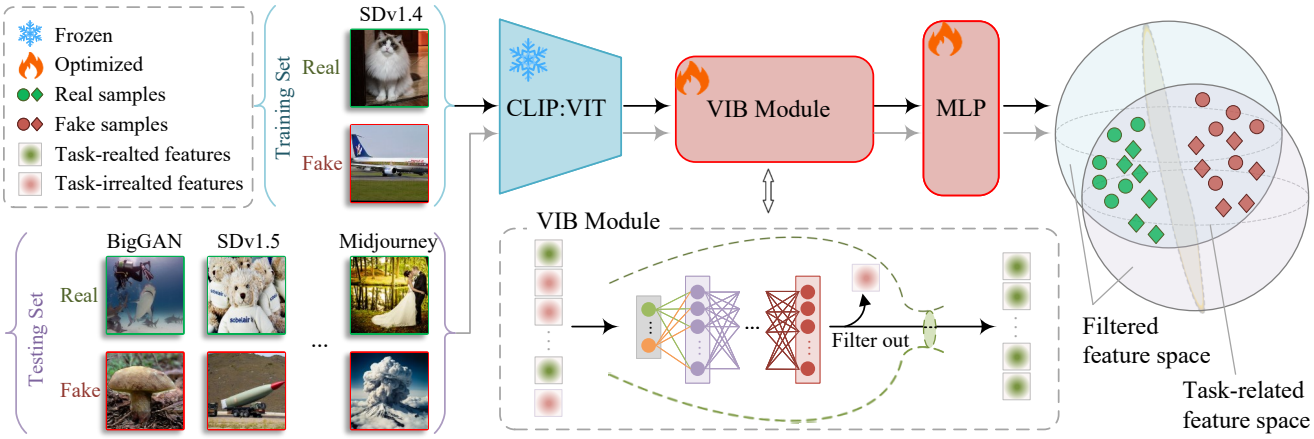


Figure 4. **Overview of VIB-Net.** We utilize Variational Information Bottleneck (VIB) techniques to map the image features extracted by CLIP's visual encoder into latent features z, where the information in z is filtered based on the Information Bottleneck (IB). Subsequently, the latent features z are used as input to an MLP to learn the decision boundary.

The encoder we use is defined as $p(z|x) = p_\theta(z|x) = \mathcal{N}\left(z|f^\mu(x), f^\Sigma(x)\right)$, where $f(.)$ represents a shallow MLP, μ denotes the K-dimensional mean vector of the output z, and Σ represents the $K \times K$ covariance matrix. We select a parameterized Gaussian distribution for both $p_\theta(z|x)$ and $r(z)$ to allow for the analytical computation of KL divergence. Additionally, we employ the reparameterization trick to estimate gradients [18]. Combining all the components above yields the following objective function:

$$L_{VIB} = \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}_{z\sim p_\theta(z|x)}\left[-\log q\left(y_n|z_n\right)\right] + \beta \mathrm{KL}\left[p_\theta\left(z|x_n\right), r(z)\right] \quad (6)$$

4.3. Workflow of the overall method

As shown in Fig. 4, the parts in VIB-Net work in the following way: First, the real and generated images $\{x_1, x_2, x_3, ..., x_n\}$ are input into the CLIP image encoder, yielding the corresponding image features $\{t_1, t_2, t_3, ..., t_n\}$. Next, these feature vectors are passed into the Variational Information Bottleneck (VIB). Specifically, the image features are processed through a shallow MLP, then a linear layer $f(.)$ calculates μ and Σ, which are transformed using a softplus activation. This layer has K dimensions, representing the bottleneck size. At this point, we obtain the latent compressed feature z, which is then fed into a linear layer $q(y|z)$ to output the target task label y.

5. Experiments

5.1. Settings

To better evaluate the model's performance, we utilize two mainstream public datasets: ForenSynths[39] and Genimage[46]. The ForenSynths dataset primarily consists of images generated by GANs, while the Genimage dataset mainly includes images generated by diffusion models. Consequently, we set up two training and testing schemes: (1) Training on ForenSynths and then testing on both ForenSynths and Genimage, and (2) similarly, training on Genimage and testing on both Genimage and ForenSynths. This setup allows us to more effectively validate the generalization performance of the detector on cross model genera scenarios. The details are as follows:

Training set:

The ForenSynths training set consists of 20 different categories, each containing 18,000 real images from the LSUN dataset [42] and an equal number of images generated using ProGAN[15]. Following the setup in Genimage, we use images generated by Stable Diffusion v1.4 [32] in the Genimage dataset, along with their corresponding real images, as the training set. In total, these include 32,000 real and generated images.

Testing set:

11 models from ForenSynths The test set includes images generated by 11 different models, primarily from the GAN family[15][45][29][5][16]. The real images are sourced from six datasets, including ImageNet[34], LSUN[42], FaceForensics++[33], CelebA-HQ[15], CelebA[22], and COCO[20].

7 models from Genimage To validate the method across different generation models, we use images generated by seven advanced generators along with their corresponding real images as part of the test set. These seven generation models include Stable Diffusion v1.4/1.5[32], VQDM[10], ADM[7], GLIDE[27], Wukong, and Midjourney.

Real-vs-Fake classification baselines: We compare VIB-Net with existing state-of-the-art detection methods, in-

Detection method	Diffusion Models							Generative Adversarial Networks						Others		*AP*
	SDV1.4	SDV1.5	ADM	GLIDE	Midjourney	Wukong	VQDM	Pro-GAN	Cycle-GAN	Big-GAN	Style-GAN	Star-GAN	Gau-GAN	Deep-fake	SAN	Average
CNNSpot[39]	99.98	99.83	51.10	58.80	67.93	99.80	49.92	53.15	50.23	49.79	55.98	47.07	56.08	54.86	54.03	63.24
Fusing[14]	99.90	97.98	69.30	94.20	81.20	99.90	84.60	67.63	87.79	69.37	67.90	91.20	43.08	72.56	89.42	81.07
Lgrad[36]	99.94	99.92	58.52	84.00	91.06	99.72	56.34	83.59	90.24	47.51	82.74	**99.19**	49.25	66.49	65.09	78.24
Univfd[28]	96.04	96.26	66.34	93.73	92.08	90.98	74.53	51.77	63.42	75.81	54.12	54.93	65.99	70.24	83.34	75.31
NPR[37]	**100.00**	**99.97**	94.70	95.80	95.50	**100.00**	86.30	83.30	94.90	72.00	82.70	97.30	66.00	**85.30**	**95.90**	89.98
CLIPping[17]	93.97	93.10	68.00	87.44	77.34	86.52	77.17	88.54	88.44	85.33	77.23	89.82	81.56	61.19	57.37	80.87
Ours	**100.00**	**99.97**	**95.49**	**97.13**	**97.81**	99.93	**97.00**	**96.59**	**98.44**	**97.17**	**84.31**	97.60	**96.94**	81.32	93.27	**95.53**

Table 1. **The *AP* values of different methods based on diffusion sources.** Models outside the Diffusion Models (DMs) column are considered as the generalization domain across generative model series.

Detection method	Diffusion Models							Generative Adversarial Networks						Others		*ACC*
	SDV1.4	SDV1.5	ADM	GLIDE	Midjourney	Wukong	VQDM	Pro-GAN	Cycle-GAN	Big-GAN	Style-GAN	Star-GAN	Gau-GAN	Deep-fake	SAN	Average
CNNSpot[39]	99.48	99.35	50.10	50.90	56.42	97.90	50.04	50.27	49.81	50.10	50.98	49.77	50.38	51.98	50.22	60.51
Fusing[14]	99.90	**99.91**	51.30	57.50	52.30	99.90	64.20	51.20	52.40	53.50	50.20	58.20	49.32	51.02	64.84	63.71
Lgrad[36]	99.12	99.05	53.00	64.24	76.34	97.53	50.93	61.61	60.74	48.82	61.43	50.17	49.70	50.17	56.49	65.29
Univfd[28]	83.55	84.80	53.35	75.30	71.60	73.55	55.10	58.65	59.30	61.45	56.80	61.45	55.30	58.40	72.00	65.37
NPR[37]	**100.00**	99.90	73.00	**89.70**	82.30	**100.00**	68.30	60.30	67.20	59.20	58.00	73.20	52.00	**74.80**	**89.60**	76.50
CLIPping[17]	96.07	95.48	70.14	85.00	77.66	88.86	79.35	88.78	88.48	89.57	**80.69**	**90.82**	**85.94**	66.91	61.64	83.03
Ours	99.55	99.20	**73.85**	74.25	**88.05**	98.25	**89.35**	**89.70**	**88.60**	**91.20**	74.10	80.70	**87.15**	72.00	81.50	**85.83**

Table 2. Similar to Tab. 1, we compare these methods using classification accuracy, which is influenced by manually set thresholds. For a fair comparison, the classification threshold for all models is set to 0.5.

cluding CNNDetection (CVPR 2020) [39], FreDect (ICML 2020) [8], Fusing (ICIP 2022) [14], LGrad (CVPR 2023) [36], DIRE (ICCV 2023) [40], Ojha (CVPR 2023) [28], NPR (CVPR 2024) [37], CLIPping(ICMR 2024) [17]. To ensure a fair comparison, we re-train and predict using the official code on ForenSynths and Genimage.

Evaluation metrics We follow existing research and consider two metrics: Average Precision (AP) and Accuracy (ACC). Average Precision (AP) is threshold-independent and evaluates the overall performance of the model. Accuracy is defined as the number of correct predictions divided by the number of test images, and this metric is influenced by the decision threshold. Given that there is no prior information in real-world scenarios, we standardize the decision threshold to 0.5.

Implementation Details Our method is implemented using the PyTorch framework on the NVIDIA Tesla V100. The model is trained with the Adam optimizer, a learning rate of 2×10^{-4}, and a batch size of 256. Following the same data augmentation strategy as in [39][28], which includes Gaussian blur and JPEG compression, experiments have shown that this is critical for the model's generalization ability.

5.2. Performance evaluation based on Diffusion sources

It is well-known that images generated by the current state-of-the-art diffusion models are more realistic than those generated by other types of generative models. To validate the effectiveness of the proposed method, we designed a challenging experimental scenario. Specifically, we trained various detector using Stable Diffusion v1.4 and then tested them on images generated by 15 different models, which include the latest diffusion-based generative models as well as classic GANs. The results are presented in Tabs. 1 and 2. Bold indicates the highest value in the column, and underline indicates the second highest value in the column. This applies to the other tables as well.

We first evaluated the performance of each model on the latest generative models. As shown in Tabs. 1 and 2, advanced diffusion models significantly challenged most detection models, causing a sharp drop in performance. For example, Univfd and Lgrad experienced a 40% decrease in accuracy and precision when tested on ADM, Midjourney, and VQDM. This decline is likely due to the high-quality images produced by these models, which make it difficult for detection methods to extract distinguishing fea-

Detection method	Generative Adversarial Networks						Perceptual loss		Others		Diffusion Models							*AP*
	Pro-GAN	Cycle-GAN	Big-GAN	Style-GAN	Star-GAN	Gau-GAN	CRN	IMLE	Deep-fake	SAN	SDV1.4	SDV1.5	ADM	GLIDE	Midjourney	Wukong	VQDM	Average
CNNSpot[39]	99.99	96.40	87.50	96.94	94.24	98.28	96.61	98.76	64.42	55.89	52.86	53.25	65.14	68.10	56.60	51.15	69.49	76.80
FreDect[8]	99.99	84.77	93.62	88.97	99.48	82.85	82.27	69.77	70.77	49.50	38.50	38.41	63.72	54.73	47.25	40.44	86.01	70.06
Fusing[14]	**100.00**	95.50	90.76	99.48	99.82	88.32	90.32	96.02	71.12	77.33	65.30	65.62	74.85	77.44	69.91	64.53	75.42	82.46
Lgrad[36]	99.90	94.01	90.75	**99.80**	**99.98**	79.29	61.64	67.05	71.71	45.09	70.90	71.72	71.83	75.96	71.42	66.51	70.23	76.93
Univfd[28]	**100.00**	99.21	98.31	97.98	99.35	99.80	**96.72**	**99.00**	82.04	82.18	85.48	82.30	84.34	84.04	69.10	90.13	94.96	90.88
NPR[37]	**100.00**	98.50	87.80	**99.80**	99.90	85.50	50.00	50.00	82.40	71.60	84.00	84.60	74.60	85.70	**85.40**	80.50	81.20	82.44
CLIPping[17]	99.85	94.03	91.27	95.24	99.05	89.52	79.87	80.33	73.89	56.62	58.28	57.85	77.25	77.93	52.43	60.65	80.00	77.89
Ours	**100.00**	**99.80**	**99.29**	98.79	99.72	**99.99**	90.89	96.83	**92.64**	**91.62**	**87.24**	**86.98**	**87.88**	**88.53**	75.68	**90.92**	**96.51**	**93.14**

Table 3. **The *AP* values of different methods trained on GAN source images.** Generative models other than those in the GANs column can be considered as the generalization domain.

Detection method	Generative Adversarial Networks						Perceptual loss		Others		Diffusion Models							*ACC*
	Pro-GAN	Cycle-GAN	Big-GAN	Style-GAN	Star-GAN	Gau-GAN	CRN	IMLE	Deep-fake	SAN	SDV1.4	SDV1.5	ADM	GLIDE	Midjourney	Wukong	VQDM	Average
CNNSpot[39]	99.99	87.59	71.18	89.95	94.60	81.44	**85.87**	85.87	51.69	50.00	50.82	50.88	60.20	57.85	50.77	51.13	56.20	69.18
FreDect[8]	99.36	78.76	81.97	78.01	94.62	80.57	60.04	60.13	63.29	50.00	40.02	40.38	64.66	55.43	46.89	41.54	78.95	65.57
Fusing[14]	99.90	87.01	77.32	85.21	97.04	76.95	85.17	**88.54**	53.76	54.56	51.04	51.34	56.50	57.15	52.12	51.67	55.09	69.43
DIRE[40]	**100.00**	76.65	72.12	85.03	**100.00**	67.79	-	-	-	-	49.74	49.83	75.78	71.75	58.01	54.46	53.68	-
Lgrad[36]	99.80	86.94	85.63	91.08	99.27	72.49	50.49	50.50	56.42	44.47	63.03	63.67	67.10	66.10	56.20	63.60	67.02	69.64
Univfd[28]	99.90	98.50	94.50	84.40	95.85	99.50	59.50	72.00	67.40	56.50	63.10	63.57	66.90	61.70	57.85	71.06	85.00	76.31
NPR[37]	99.80	96.10	84.40	**97.70**	99.30	82.50	50.00	50.00	80.20	69.20	**76.60**	**77.90**	69.70	77.30	**77.80**	**76.10**	78.10	78.98
CLIPping[17]	99.88	96.74	94.77	94.87	99.47	94.15	85.23	87.73	76.48	60.27	61.26	60.78	**80.77**	**80.69**	53.95	63.94	84.76	80.93
Ours	99.99	**99.00**	**95.75**	91.25	98.95	**99.70**	71.25	86.75	**83.20**	**70.50**	71.55	70.00	71.45	69.40	61.25	75.90	**86.65**	**82.50**

Table 4. ***Accuracy* of different methods trained on GAN source images.** The data from [40] is obtained from [37], and "-" indicates that the method did not show data for the corresponding model.

tures. We also observed that these models performed better on generative models with distributions similar to the training data (e.g., SDV1.5, Wukong). This suggests that they may have learned superficial biases from irrelevant features, which are superficially correlated with labels and do not generalize well to images from different distributions. In contrast, NPR, which detects authenticity by analyzing pixel relationships, partially mitigates these biases, resulting in less performance decline. When testing on GANs, we observed significant performance drops, particularly on GauGAN, where the accuracy and precision were around 50-60%. This further highlights how even subtle relevant features and surface biases can severely impact the generalization performance of these models.

In contrast, Our method outperforms the baseline model, Univfd, with improvements of 20.22% in average precision and 20.46% in accuracy. This gap is even more pronounced across generative model families (GANs), where our method surpasses Univfd by 34.17% in precision and 26.41% in accuracy. Compared to state-of-the-art methods, our approach improves average precision by 5.55% and accuracy by 9.33%. For other generative models (GANs), VIB-Net achieves gains of 12.48% in average precision and 23.59% in accuracy. This is due to VIB-Net's ability to retain compact deep features while eliminating irrelevant surface features, enhancing generalization. Overall, these results highlight the advantages of our method.

5.3. Performance evaluation based on GAN sources

So far, we have explored the generalization performance of using images generated by specific diffusion models as training data. Next, we trained using images generated by ProGAN and then tested on 18 different generative models to explore the performance of the detector on unseen generative models. The results are shown in Tabs. 3 and 4. From the tables, we can see that almost all models perform well when detecting images generated by other GAN variants. This is because, compared to diffusion-based generation methods, images generated by these GAN variants are visually more distinguishable, making it easier for the model to learn the distinguishing features of real and fake. However, when detecting images generated by models other than GAN variants, the performance of most models sharply declines. This suggests that even when the training data contains rich features related to real/fake classification, it does not necessarily ensure better generalization performance. This is because the model may overfit the incidental correla-

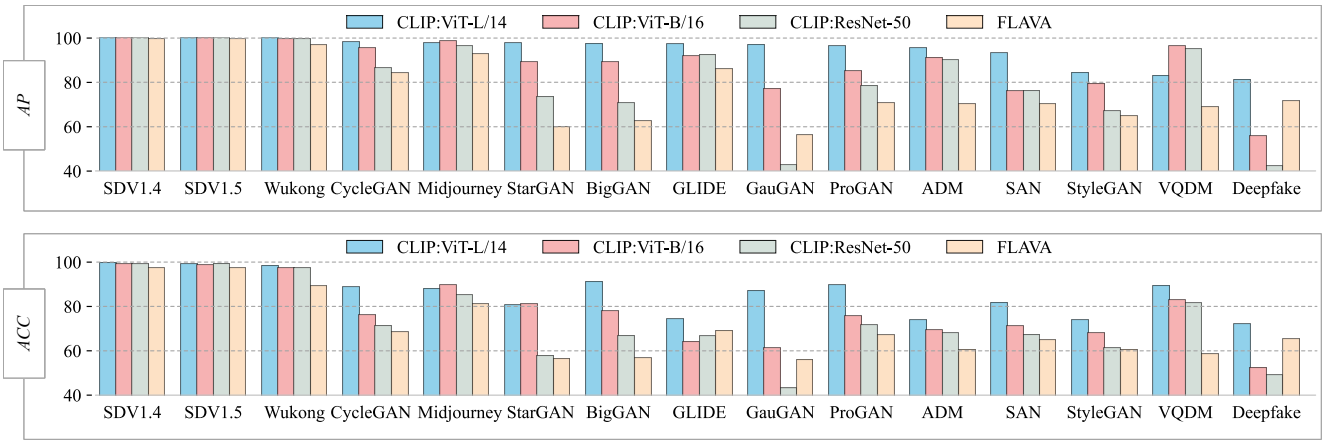


Figure 5. **Ablation study on the network architecture.** The *AP* and *ACC* values of different VIB-Net variants on various models. CLIP:ViT-L/14 shows better performance.

tions between irrelevant features and the task, which harms generalization performance. Univfd performs better across model families, but since it extracts general features, many of these features are unrelated to the real/fake classification task, which naturally affects the decision boundary.

In contrast, our method shows better performance in detecting real and fake images. Compared to the baseline Univfd, our method achieved improvements of 2.26% and 6.19% in average precision and accuracy. In OOD scenarios (excluding GANs), our method outperforms the baseline Univfd by 3.22% and 8.48%. Compared with advanced methods, our method achieved improvements of 10.7% to 23.08% in average precision. These results further validate the effectiveness of our proposed method.

5.4. Effect of VIB-Net across different backbone

So far, we have seen the impressive performance of VIB-Net with the CLIP:ViT-L/14 backbone. In this section, we will explore the effect of transferring the VIB component to other multimodal models. We consider the following backbone variants to form VIB-Net: (1) FLAVA-FULL[35] (2) CLIP:ResNet50 (3) CLIP:ViT-B/16 (4) CLIP:ViT-L/14[30]. For each VIB-Net variant, we train using images generated by Stable Diffusion v1.4 and corresponding real images. Fig. 5 show the AP and ACC results of these variants on the same generative model. We observe that among these multimodal models, ViT-L/14 achieves the best performance, which can be attributed to its larger architecture and attention-based global receptive field. More importantly, all these variants exhibit strong performance, which can be attributed to the unique capabilities of our method.

5.5. Analysis of latent features

Finally, we apply our method to address the issues analyzed in Sec. 3.2. In Sec. 3.2, we identified that some irrelevant features in CLIP's general features could affect the classifier's decision boundary, causing the model to learn superficial biases, which in turn impacts its generalization performance. As a solution, we proposed VIB-Net. Following the same experimental setup as in Sec. 3.2, we visualized the latent features extracted by VIB-Net using UMAP, and obtained encouraging results, as shown in Fig. 3b. We found that the latent features no longer exhibit obvious irrelevant features related to the real/fake image classification task, as the two separated clusters now belong to the same category (real or generated), unlike in Fig. 3a, where features of the same category were significantly separated. This demonstrates the advantage of our method.

6. Conclusion

We propose a universal real-fake image detection method, VIB-Net. First, we found that the general features extracted by current methods based on large-scale pre-trained models contain irrelevant features that are unrelated to the task of distinguishing real from fake images. Experiments have shown that these features not only affect classification decisions, but may also cause the model to learn superficial biases, thereby damaging its generalization performance. As an alternative, we use VIB to remove task-irrelevant features, aiming to obtain the most concise image representations. These simplified features are maximally correlated with the real-fake classification task.

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